Anurag Pradhan

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Portfolio | LinkedIn | GitHub

Summary

Prefinal-year Computer Science student at VIT Chennai with hands-on experience in Deep Learning and Machine Learning. Skilled in building and deploying efficient CNN, Transformer, and Vision-Language models, along with PyTorch experimentation, data preprocessing, optimization, and on-device inference. Published researcher (IJC-NLP–AACL 2025) focusing on accessibility-centered evaluation of lightweight VLMs. Strong data science foundation, including feature engineering, exploratory analysis, and supervised learning. Adept at delivering practical, production-ready ML solutions within engineering teams.

Education

- **Vellore Institute of Technology** — Chennai, India
  Bachelor of Technology in Computer Science; GPA: 7.74/10 — *Aug 2023 – June 2027*
- **Sri Venkateswara Jr. College** — Visakhapatnam, India
  Class 12 (MPC); Score: 88.4/100 — *July 2021 – June 2023*
- **Sri Chaitanya Techno School** — Visakhapatnam, India
  Class 10; Score: 93.6/100 — *Feb 2020 – Mar 2021*

Skills Summary

- **Programming Languages**: Python, C, C++, Java
- **Core Expertise**: Deep Learning, Computer Vision, CNNs, Vision Transformers, Image Processing, Sequence Models (LSTM/GRU/RNN), Generative AI, Large Language Models
- **Frameworks & Libraries**: PyTorch, Hugging Face Transformers, LLaMA, BEiT, BLIP, Llama.cpp
- **Tools & Platforms**: Git, Docker, CUDA, ffmpeg, Linux
- **Soft Skills**: Leadership, Technical Writing, Event Management, Time Management

Experience

- **National Aluminium Company Ltd (NALCO)** — Damanjodi, Odisha
  Computer Science Intern — *June 2025 – July 2025*
  - **Contribution**: Worked on Respiguard AI, an AI-based respiratory safety system for mining environments. Improved hazard detection reliability using deep-learning-based visual analysis.
  - **Tech Stack**: Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation
- **Team Ignition, VIT Chennai** — On Site
  Flight Software Lead — *Apr 2024 – Present*
  - **Contribution**: Developed flight software for model rockets, including fin actuator control, stability algorithms, onboard video stabilization, and Kalman-based state estimation.
  - **Tech Stack**: Arduino, C/C++, PID Control, Image Processing, Deep Learning

Projects

- **Liver Disease Prediction**: Developed a YOLOv5-based detection framework with Stable Loss, augmentation strategies, and Adam optimization. Added interpretability through heatmap visualizations. **[GitHub Repo]**
  Tech: Python, PyTorch, YOLOv5, CUDA
- **Brain Tumor Segmentation**: Built a BEiT-transformer segmentation pipeline with skip connections and attention-driven decoding. Trained with Albumentations and CE–Dice hybrid loss for stable convergence. **[GitHub Repo]**
  Tech: Python, PyTorch, Vision Transformers, Semantic Segmentation
- **JanaSathi: Odia E-Governance Chatbot**: Designed a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings. Integrated Odia translation with Cohere Command-R+. Achieved 2nd place in AMD-sponsored OdiaGenAI Hackathon 2025. **[GitHub Repo]**
  Tech: Python, Transformers, LangChain, Retrieval-Augmented Generation
- **Mobile Deployment of Small Vision-Language Models**: Deployed SmolVLM2 (2.2B & 500M) Video-Instruct models on-device (Vivo Y27) using llama.cpp with FP16 and Int8 optimization. Demonstrated real-time video captioning within strict mobile constraints. **[GitHub Repo]**
  Tech: VLMs, Quantization, Llama.cpp, ffmpeg, Termux, Kotlin

Conference Publications

- **Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-Evals**: S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP–AACL 2025), Mumbai, India.